

Data Science Program

Capstone Report - Spring 2026

CTRL+C-YA (Instagram Impersonated Accounts Detection) Capstone Project

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A capstone presented in fulfillment of the requirements for the degree of Bachelor's of Science in Data Science

supervised by
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Preface

As a data science student at George Washington University, I have spent my academic years developing skills in data collection, mining, visualization, ethics, and applying machine learning to solve digital world problems. The CTRL+C-YA project was inspired by my own experience, frustration with social media impersonation, and deep interest in data and cybersecurity. I developed the web tool to offer everyday users and cyber professionals an automated way to protect individuals' digital identities before damage occurs to them. By sharing these findings, I hope to provide a practical framework that helps both Instagram's users feel safe and cyber experts innovate new ways to verify assets in the hyper-connected world.

Acknowledgements

I am deeply grateful to Professor Sarah Burnett for her patience and expertise in supporting the project process. Also, big thanks to George Washington University's Data Science Department for the resources and opportunity, and to my classmates for the honest feedback that kept me on track.

Abstract

Social media impersonation is an increasing threat to social media accounts' security. Impersonators can easily obtain personal data and use it to spread misinformation or defraud other users on Instagram for their advantage. The issue had been very difficult to tackle because of low barriers to account creation, allowing impersonators to generate new profiles faster than platforms can manually control. To address this, my research analyzes data scraped from sources like InsTrack and ViralPitch, using machine learning models to identify key features and behavior patterns associated with impersonators. I found that a Random Forest model provided the highest accuracy, primarily by leveraging the followers-to-following ratio, an effective feature for impersonator accounts, which often follow many users aggressively while having fewer followers themselves. These findings are integrated into the CTRL+C-YA web tool, which provides users with a quick, data-driven way to detect impersonator accounts and stay protected before an impersonation process occurs.

Keywords

Social Media; Cybersecurity; Impersonation Detection; Feature Engineering; Impersonated Account; Instagram; Machine Learning Model; Capstone Project

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1 Introduction

In 2025, Meta removed 10 million impersonated accounts from platforms like Instagram to tackle spam and restore users' trust.[1] These accounts replicate a real individual's entire posts and personal biographies to conduct fraudulent activities by exploiting a user's personal network. While they may appear harmless, like simple fan accounts, they pose a severe threat. One scheme alone can defraud 1,400 users for $2.8 million.[2] Currently, statistics show that 30% of all fraud victims are targeted through social media. [3] Beyond financial loss, impersonation inflicts lasting reputation damage. Famous figures, such as Sir Tom Jones, frequently have to warn fans about fraudulent accounts soliciting money in their name.[4] These impersonations destroy an individual's reputation that they take years to build. Since manual reporting and platform sweeps cannot keep pace with this volume, there is a clear need for automated, accurate detection tools for impersonated accounts.

This occurring issue motivated the development of CTRL+C-YA, an interactive web tool designed to detect Instagram impersonators. Unlike passive reporting systems, CTRL+C-YA uses a survey-style interface where users input their profile details, including username, name, and bio, to search for similar accounts. The system will filter and sort out the top five accounts in the web database that identify matches with 80% similarity or higher to the user input. To ensure high accuracy, the tool utilizes a cleaned dataset that filters for suspicious behavioral patterns. Key features, especially the Followers-to-Following ratio, were rigorously tested and trained in machine learning models like Random Forest before being integrated into the web interface. By combining profile similarity percentages with behavioral metrics, the tool generates an overall score to help users efficiently identify and clear out any potential impersonate account.

2 Related Work

Recent efforts in social media security have prioritized the detection of Instagram impersonation. A study from the National Technical University of Athens shows that selecting fewer features and a low amount of data, such as followers-to-post ratio, often outperforms massive datasets. [1] Supporting this focus on efficiency, researchers have demonstrated that a bagging classifier utilizing a specific subset of profile features, such as username length, can correctly classify over 98% of account with a low error rate. [2] Building on this, another study utilizes hybrid machine learning, like Random Forest and XGBoost, while

[1]"Meta Removes 135,000 Instagram Accounts Targeting Children in 2025 Clampdown." 2025. TradingView. July 23, 2025.

[2]"Hidden Dangers of Fake Instagram Profiles You Must Know | Breaking AC." 2025. Breaking AC. December 8, 2025. https://breakingac.com/news/2025/dec/08/hidden-dangers-of-fake-instagram-profiles-you-must-know/.

[3]"New FTC Data Show People Have Lost Billions to Social Media Scams." 2026. Federal Trade Commission. April 27, 2026. https://www.ftc.gov/news-events/news/press-releases/2026/04/new-ftc-data-show-people-have-lost-billions-social-media-scams.

[4]McCarthy, James. 2026. "Sir Tom Jones Warns of Fake Profiles Impersonating Him on Instagram," January 9, 2026. https://www.bbc.com/news/articles/c394v84z9lko.

applying SMOTE and GridSearchCV to fix imbalanced data and refine model reliability. [3] Additionally, prior researchers even prioritize transparency by using SHAP values to explain the reason an account was flagged, while others move away from traditional machine learning entirely to use deep learning models like LSTM for high-dimensional raw data. [4][5] These various methodologies (feature optimization, ensemble learning, and interpretability) form the baseline for identifying the shifting patterns of impersonators. While this previous research provides a strong foundation, my research project differs by shifting the focus from static experimental accuracy to realistic, real-world usability. Most existing research concludes with advanced models in a controlled environment. However, the CTRL+C-YA project transforms those models into a functional, interactive web-based tool for immediate user use. Unlike the heavy resource neural networks found in recent literature reviews, this project demonstrates that efficient detection is possible through a streamlined Random Forest approach that prioritizes the most effective behavioral identifiers. Furthermore, where many studies rely on stale, public datasets, CTRL+C-YA utilizes the most recent data to account for the consistent change of impersonator approaches. By combining ethical data handling with a simplified user interface, the project bridges the gap between academic theory and a practical defense against modern digital identity theft.

3 Solution

The project follows a structured pipeline to ensure accurate detection and high performance. This pipeline consists of five phases: data acquisition, data cleaning, feature engineering, model, and web development. Also, it covers the pipeline of the final CTRL+C-YA web operations. This flow ensures that the data is prepared fairly and produces advanced results for real-world applications.

3.1 Data Acquisition

To identify the behavior of impersonated accounts, I collected profile information specifically from Instagram. The raw data consists of an existing dataset from GitHub combined with web-scraped data from sources like Instrack, ViralPitch, and Hopper HQ. I used Selenium to scrape these sites because it avoids URL blockages by reflecting human browsing. The script parses HTML and converts it into DataFrames, capturing account data ranging from 2016 to 2026. This variety helps identify significant features even with a smaller dataset.

3.2 Data Cleaning

During this phase, I clustered the information into two CSV files: the fake or impersonated accounts and the real accounts. Using Pandas and NumPy, I converted string numbers to digits, filled missing values with “NA,” and removed duplicates based on the unique ID that was assigned to each distinct account. I

renamed core columns to username, number of posts, followers count, and bio for consistency. Based on research and literature review, I added calculation columns like bio lengths, number of digits in username, and followers and following ratio.

3.3 Feature Engineering

Before finding significant features, I created a few visualization plots using Matplotlib and Seaborn to visualize account behaviors and avoid bias. In Figure 1, the bar graph counts the number of users between fake accounts and real accounts within the combined dataset. This confirms that the dataset is balanced between the real and fake accounts to ensure the models do not favor one class. For Figure 2, the scatter plot presents the relationship between the followers and following counts based on the type of account it is and each type of account behavior when it is clustered in a certain part of the graph. The use of a scatter plot shows how followers and following counts cluster differently on impersonate and real accounts. The heatmap in Figure 3 utilizes a correlation heatmap to identify redundant features. A score of 0.95 and a dark blue color indicate features that provide nearly identical information, allowing me to narrow my focus to the most significant users.

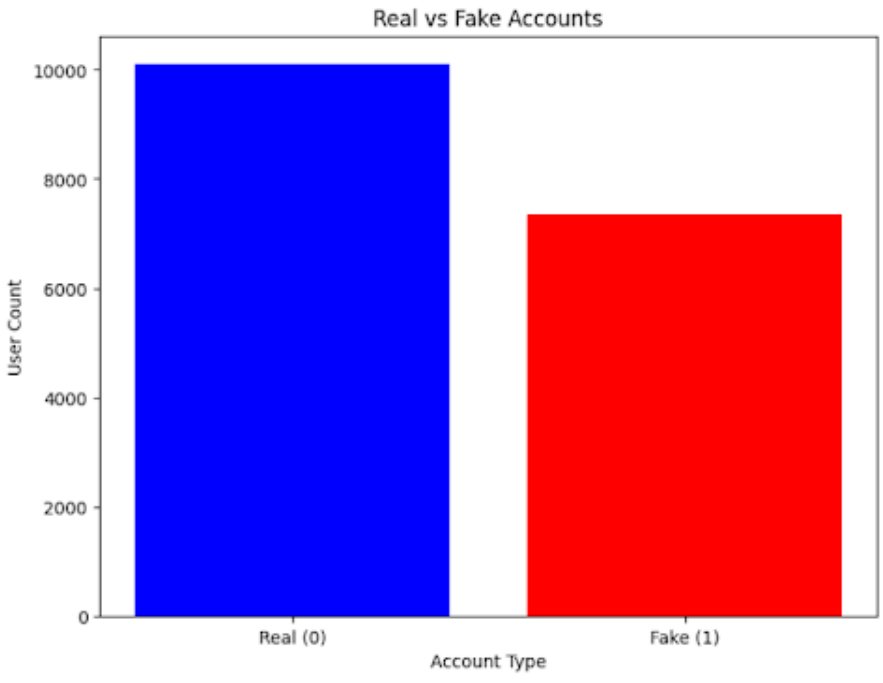


Figure 1: Real vs. Fake Users Accounts Bar Graph.

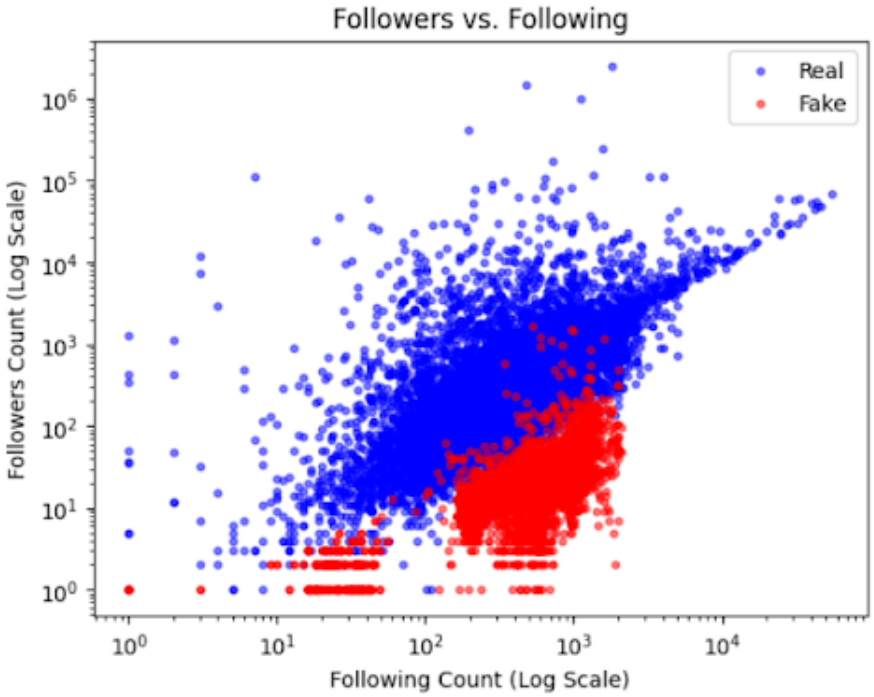


Figure 2: Relationship between Followers and Following Diverse Account Type Scatter Plots.

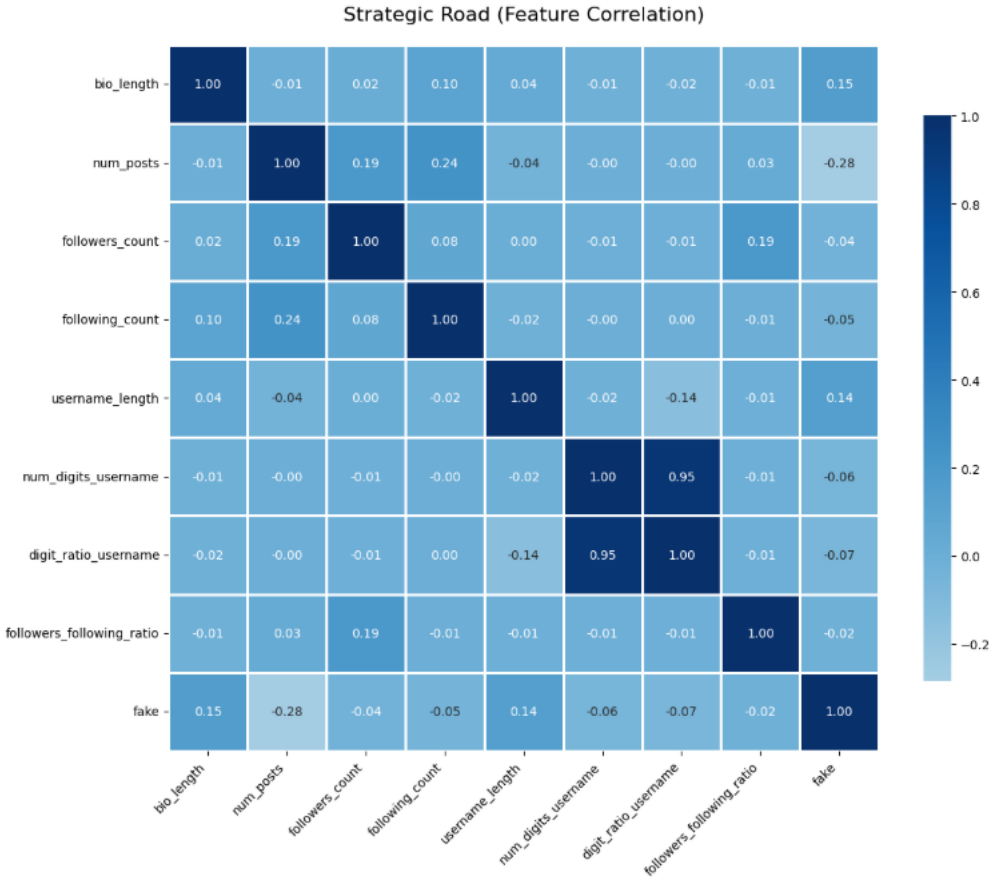


Figure 3: Dataset Features Heatmap.

3.4 Modeling

I applied a train-test split to evaluate the three machine learning models: Logistic Regression, Support Vector Classifier (SVC), and Random Forest. These models are ideal for binary classification on social media. While Logistic Regression and SVC serves as a baseline, Random Forest excels at capturing non-linear patterns in user metrics. After testing, I analyzed the importance scores to see which features impacted accuracy most. Box plots were then used to visualize these significant features from top machine learning models to differentiate impersonators from real accounts.

3.5 Web Development

The final phase involved building a functional web of CTRL+C-YA using Flask for the backend. The Python file, app.py, manages routes between the input form, the analysis engine, and the result pages. The backend uses the library SequenceMatcher to calculate the similarity percentage between user input and the database. The system also applies the patterns identified by the Random Forest model, resulting in high accuracy in flagging suspicious accounts.

To ensure high performance without the web crashing, the result page only displays the top five matches with at least an 80% similarity score. An overall score is calculated by averaging the similarity of the name, username, and bio. Accounts with an overall score of 50% or higher are flagged as potential threats if they match known fake patterns. The frontend, built with HTML and CSS, shown in Figures 4, 5, and 6, provided an interface or UI where users can easily input profile details and receive an immediate result of similarity to the user input if it is impersonated or not. Overall, this will help bring out the result of the research and let privileged users use and have more control over their own data.



Figure 4: The image shows the form page that the user first sees when coming into the website. It allows users to input information for their certain account to detect any other account that impersonates them.

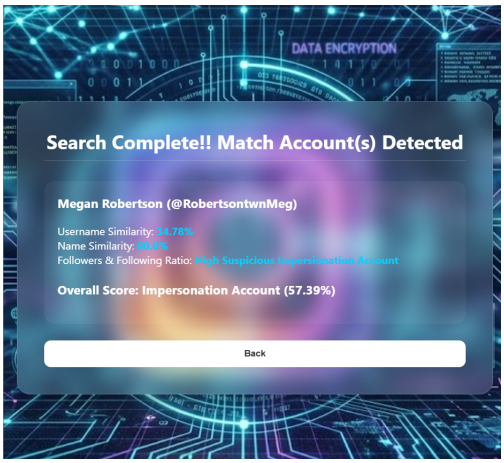


Figure 5: The result page is showing an account that detects similarity with the user's input, showing the percentage similarity of username and name, overall score, while using the significant feature to identify the followers-to-follower ratio feature from the Random Forest machine learning model accuracy.

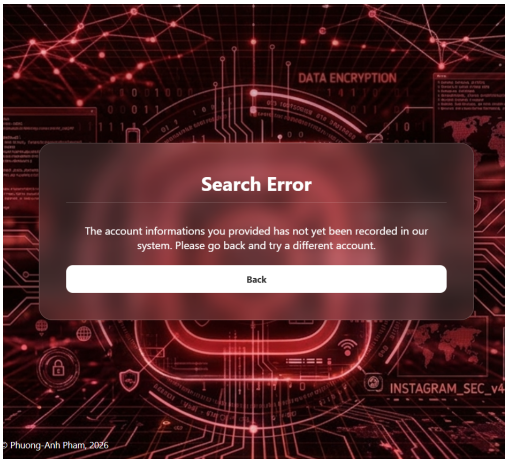


Figure 6: The Error page is where the machine cannot find a similarity from the user input in the accounts listed in the web database.

The operational process of the CRTL+C-YA system, shown in Figure 7, connects the research findings to the user experience. The pipeline begins with the user's input, moves through the similarity formula and Random Forest model patterns, and ends with a clear output of the impersonation detection result, including the similarity percentage of the features. This system empowers users to take control of their personal data and to immediately identify impersonators' accounts.

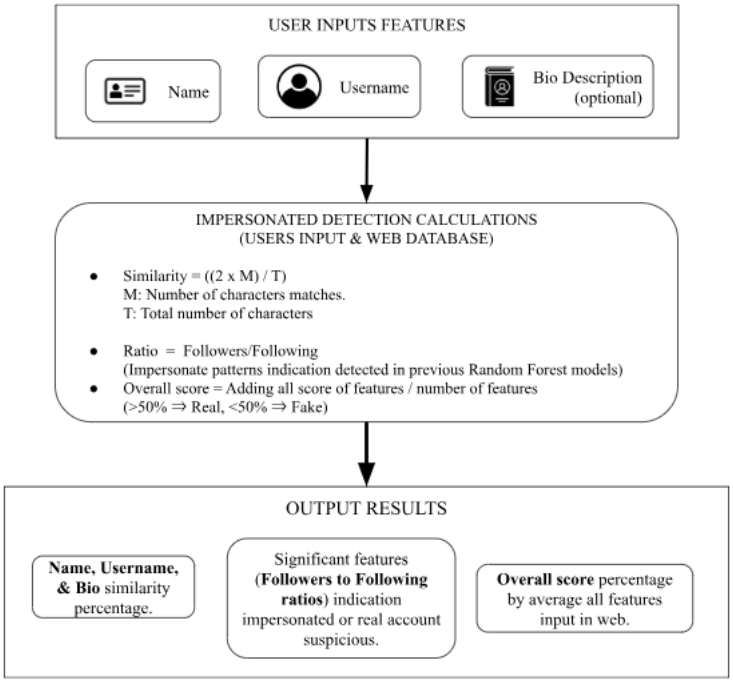


Figure 7: Web Operation Pipeline

4 Results and Discussion

This part presents the findings of the accuracy tests and validates the project's objective. The experiments identify the most effective machine learning models for detecting impersonation and highlight the key features that drive these predictions. By identifying these patterns, we can better understand the behavior of imposter accounts and integrate these indicators into the web tool to help users identify threats more quickly.

4.1 Experimentation protocol

To evaluate our solution, we tested three distinct models, Logistic Regression (LR), Support Vector Classifier (SVC), and Random Forest (RF), using a merged dataset of real and fake Instagram account data. We measured success based on machine learning model classification accuracy, confusion matrices for error analysis, and feature importance scores to see which profile traits actually significant.

4.2 Accuracy Models

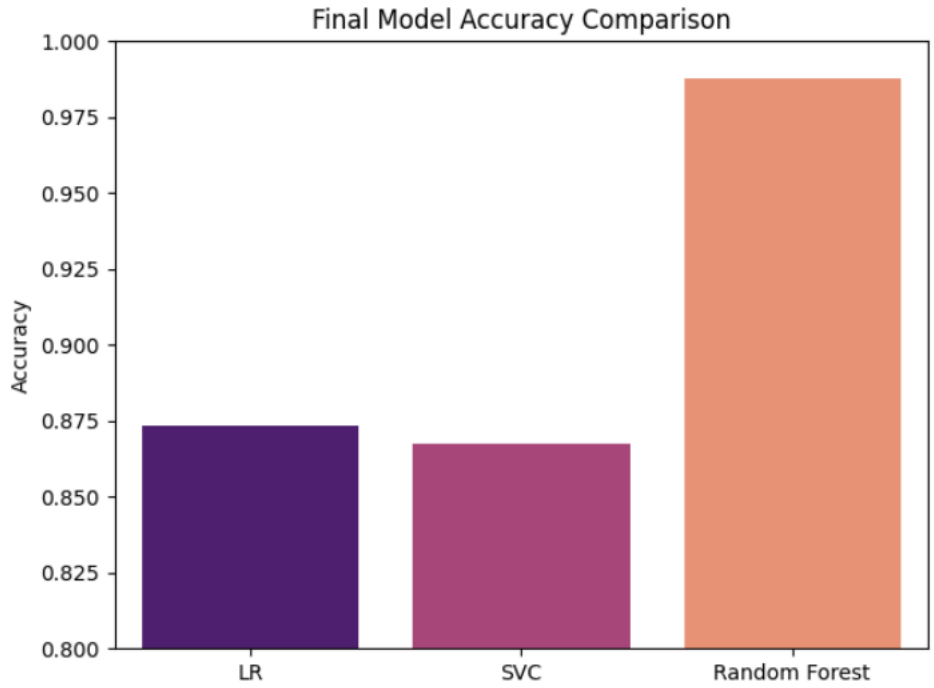


Figure 8: The bar graph shows the accuracy percentage of the three machine learning models (Logistic Regression, Support Vector Classification, and Random Forest).

Figure 8 compares the accuracy of the three models. The Random Forest model reached a peak accuracy of over 97.5%. In contrast, Logistic Regression scored roughly 87.5%, and SVC performed the lowest at approximately 86%. While LR and SVC have high accuracy, they struggle with the complex, non-linear patterns found in the dataset. Random Forest significantly outperforms them because it can handle the intricate relationship within the dataset more effectively than the straight-line boundaries used by linear models like LR and SVC.

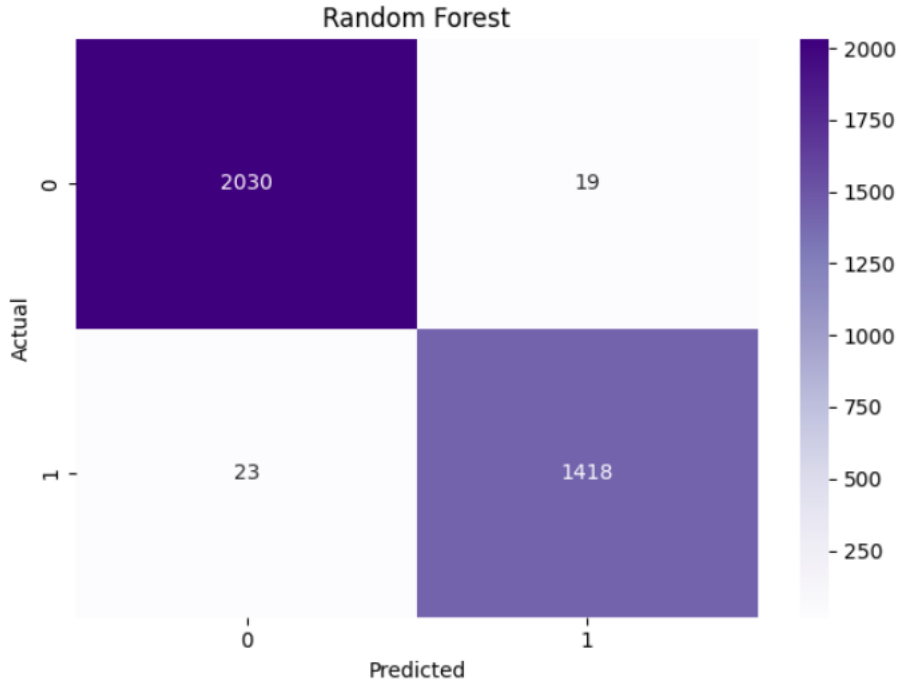


Figure 9: The confusion matrix visualizes the performance of a Random Forest classification model, illustrating its high degree of accuracy in predicting two distinct classes (0 and 1).

The confusion matrix plots in Figure 9 demonstrate how the Random Forest model performs during the test. It successfully distinguished between account types by correctly identifying 2,030 real users and 1,418 fake users. The error rate was remarkably low, with only 19 false positives and 23 false negatives. This results in a precision of 98.7% and a recall of 98.4%. Essentially, when the model flags an account as fake, it is almost always right. The clear diagonal alignment in the matrix proves that the model has established firm boundaries between the two classes.

4.3 Significant Features And It Patterns

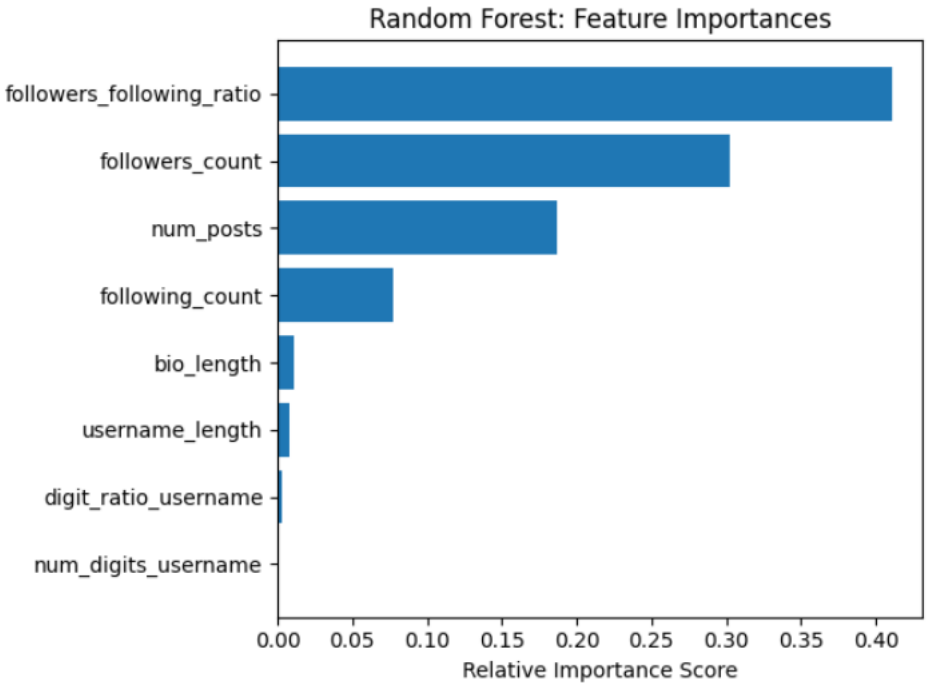


Figure 10: The horizontal bar chart displays the feature importances for a Random Forest model.

Figure 10 describes the features that influence the model the most based on the importance score. The followers-to-following ratio emerged as the most critical predictor, with an importance score over 0.40. This was followed by total follower counts and post frequency. Interestingly, the data description features like bio length and username digits had almost no impact on the model's final decision. This suggests that while similarity in names is important for humans, the machine learning model relies heavily on social growth metrics to confirm impersonation.

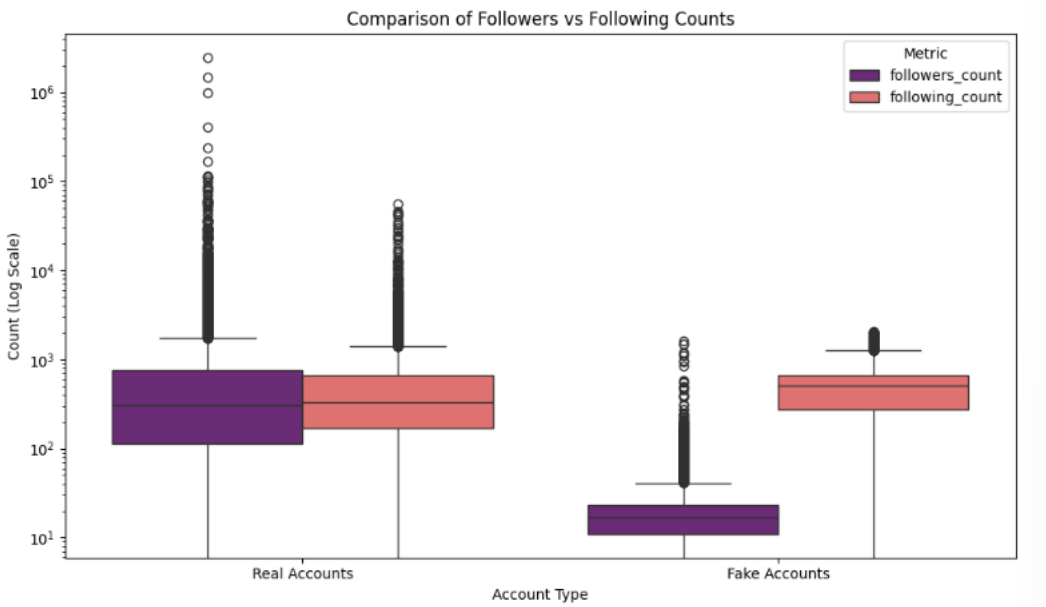


Figure 11: The log scale box plot compares the followers and following counts of real versus fake social media accounts.

The log-scale box plot in Figure 10 points out why the Random Forest model is so successful. Real accounts show a massive range of followers, often reaching around 10^3, while fake accounts are compressed below 10^2. Fake accounts also show a consistent following bias, where they follow many more individuals than the number that follows them back. Since there is virtually no overlap in follower counts between the two groups, the mode can easily separate them. Furthermore, the Random Forest ensemble is naturally resistant to the many outliers (the dots in the plots) that often cause linear models like LR or SVC to fail in this project. Overall, the research proves that impersonation follows a predictable pattern. Real accounts maintain a balanced social ratio, whereas fake accounts show a widening disparity between their following and follower counts. This specific behavior serves as a definitive red flag for detecting impersonation activity.

5 Discussion

The primary challenge in this project was web-scraping data from sources like Instagram. Privacy settings, HTML blocks, and API limits acted as significant hurdles to building a large database for research. This scarcity of data is a major issue because it can force a research focus onto a narrow cluster of features, potentially leading to bias. Another specific challenge involves the followers-to-following ratio. While useful, this metric alone is a biased indicator. Without deep cross-checking, relying on this single feature causes the tool to misclassify new user accounts, fan accounts, or business accounts that naturally have unbalanced follower and following counts. Compared to the related work that uses complex ensemble learning or deep learning , my approach is more direct and accessible for a front-end tool. It excels

at providing immediate, understandable similarity scores for social media users. However, the machine learning models mentioned in my research, like Random Forest, outperform my current logic in handling outliers. Those models can identify patterns across dozens of variables simultaneously, whereas my tool currently focuses on a few key metadata points. To mitigate these limitations, I recommend moving toward a multi-dimensional analysis. Instead of just looking at static numerical ratios, future versions should integrate behavioral patterns over time and metadata like username digit ratios to ensure a more robust, unbiased system. The current research is also limited by the small dataset, which restricts the exploration of more diverse features. Additionally, impersonators are clever; they use adversarial tactics like buying fake followers to gain legitimacy or using AI-generated profile pictures to mimic real humans. They can even stay dormant for long periods to avoid a triggering spam alert. To overcome this, a great next step would be to collaborate with platforms like Facebook or TikTok to access larger, anonymized public datasets. This would allow for real-time updates to the detection mechanism as impersonator tactics evolve. Beyond this, I feel it is worth exploring image recognition AI to flag deepfake profile pictures or natural language processing to analyze the tone of a bio, as these behavioral markers are much harder for a bot to fake than simple profile description numbers.

6 Conclusion

The CTRL+C-YA web successfully addresses the challenge of identifying Instagram impersonators by focusing on high-impact profile features. Through machine learning analysis, specifically the Random Forest model, we determined that the Followers-to-Following ratio is the most reliable indicator of fraudulent activity. While these behavioral patterns are often difficult for a person to spot at a glance, our model processes these metrics to flag any high-risk accounts with high accuracy in testing. The integration of fuzzy string matching for names and bios further strengthened the system, creating a web-based scanner that provides users with immediate, data-driven similarity scores. Looking ahead, this research provides a foundation for border social media security. To improve detection, future versions of the research should move beyond metadata and incorporate deep learning to analyze profile pictures or post content. Expanding the tool's reach to platforms like AliExpress or any online shopping would also allow us to compare how impersonation tactics differ across different social ecosystems. While the current version effectively uses a streamlined set of features, adding more diverse data points will lead to even more robust decision-making and better protection for online identities.

A Appendix

The following appendix includes supplemental figures that were not mentioned in the main report, but are important to acknowledge in this research.

Logistic Regression Accuracy: 0.8736389684813753

	precision	recall	f1-score	support
0	0.96	0.82	0.88	2049
1	0.79	0.95	0.86	1441
accuracy			0.87	3490
macro avg	0.87	0.89	0.87	3490
weighted avg	0.89	0.87	0.87	3490

Figure 12: Performance metrics for Logistic Regression classifier, including precision, recall, and F1-score.

SVC Accuracy: 0.867621776504298

	precision	recall	f1-score	support
0	0.95	0.82	0.88	2049
1	0.79	0.93	0.85	1441
accuracy			0.87	3490
macro avg	0.87	0.88	0.87	3490
weighted avg	0.88	0.87	0.87	3490

Figure 13: Performance metric for the Support Vector Classifier (SVC).

Random Forest Accuracy: 0.9879656160458453

	precision	recall	f1-score	support
0	0.99	0.99	0.99	2049
1	0.99	0.98	0.99	1441
accuracy			0.99	3490
macro avg	0.99	0.99	0.99	3490
weighted avg	0.99	0.99	0.99	3490

Figure 14: Performance metric for the Random Forest classifier.

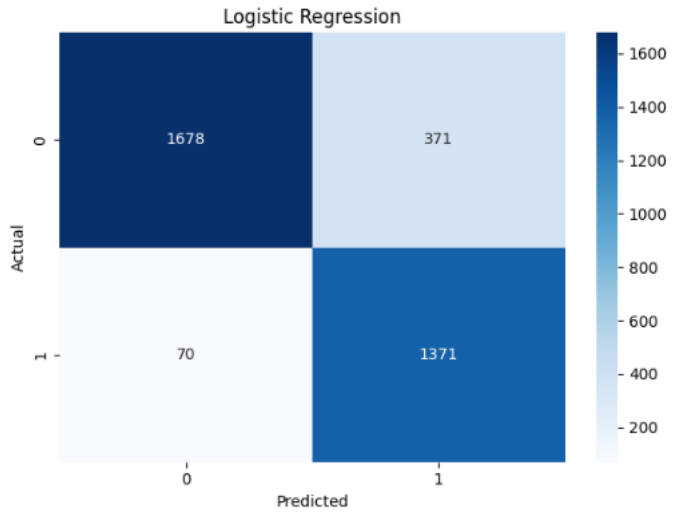


Figure 15: Confusion matrix for Logistic Regression model on the test dataset, achieving accuracy 87% ((1678+1371) / (1678+371+70+1371)).

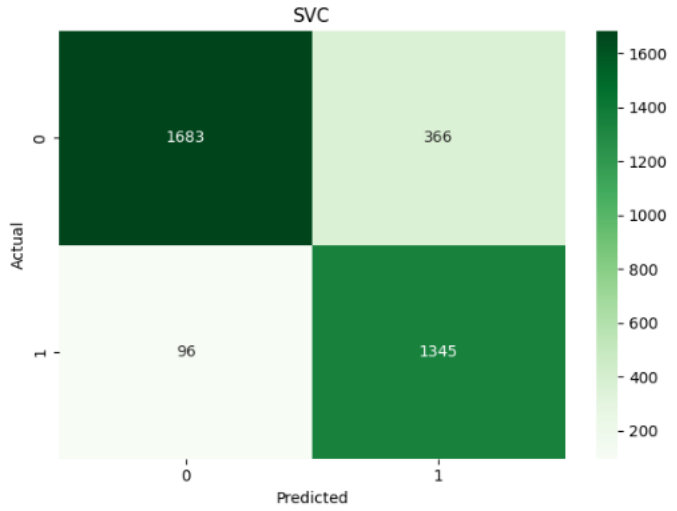


Figure 16: Confusion matrix for Support Vector Classifier (SVC) of inner side text score based on result of 87% ((1683+1345) / (1683+366+96+1345)) percent accuracy.

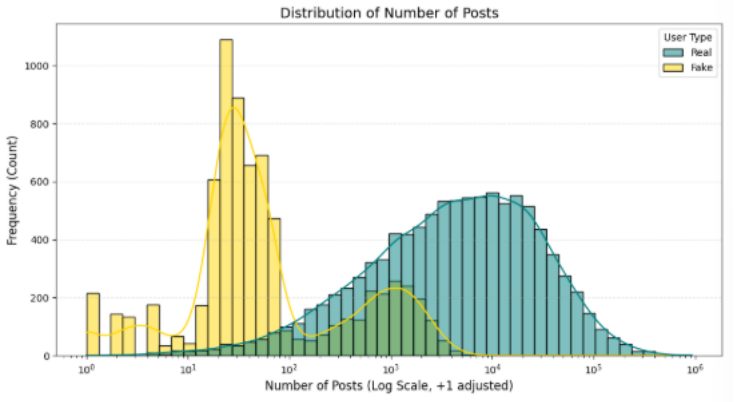


Figure 17: Distribution of the number posts for real and fake users, presented on a transformed log scale with a +1 adjustment to illustrate the frequency variation across different user categories.

References

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